

ANKIT RAJ

+91-7484083083 | ankitforward47@gmail.com | linkedin.com/in/ankitraj0714 | github.com/AR0714

TECHNICAL SKILLS

Languages: Python, C/C++ (Embedded/Arduino), MATLAB, JavaScript
ML/DL Frameworks: TensorFlow, Keras, PyTorch, TensorFlow Lite Micro (TinyML), scikit-learn
ML/DL Concepts: ANN, RNN, CNN, NLP, Vector Embeddings, Neural Network Optimization, Edge Deployment
GenAI: RAG, LLM Application Development (LangChain, HuggingFace)
Libraries & Tools: NumPy, Pandas, Matplotlib, OpenCV, Git
Embedded & IoT: ESP32, Arduino, Sensor Integration, FFT & Signal Processing

SUMMARY

- Final-year **Electrical Engineering** student working at the intersection of power systems and edge AI — building on-device ML for fault diagnosis, predictive maintenance, and condition monitoring. Recent work includes an explainable ML pipeline for transformer fault diagnosis (DGA) and a patent-pending TinyML motor-fault system.

EXPERIENCE

Summer Intern – DRDO, PXE Imaging (Chandipur)

May 2026 – Jun 2026

Python, PyTorch, OpenCV, MATLAB

- Built and benchmarked a **CNN-based stereo depth estimation** pipeline in **PyTorch** against a classical **SGBM** baseline, cutting error by **43.8%** (MAE 50.97 vs. 90.66 px) on the Middlebury dataset.
- Ran a systematic **robustness evaluation** across **21 degraded imaging scenarios** (fog, low light, noise, blur, rain), where the **CNN** held **46–51% higher accuracy** than the classical method.

PROJECTS

IoT-Enabled Edge-AI System for Predictive Maintenance | *Patent Pending*

2025

Edge Impulse (TinyML), TensorFlow Lite Micro, ESP32, Arduino/C++, Node.js/Express, React/TypeScript

- Built an end-to-end TinyML system running an on-device **healthy/faulty** motor classifier on an **ESP32** (trained in **Edge Impulse**, deployed as **TensorFlow Lite Micro**) from current and dual-axis vibration in 1-second, 100 Hz windows; achieved **≈85% accuracy** (F1 0.89) at **sub-100 ms** on-device inference.
- Streamed on-device inference via a **Node.js/Express** backend (device-key auth, Server-Sent Events) to a live **React/TypeScript** dashboard, sending compact JSON scores instead of raw sensor streams for **≈99% lower bandwidth**; filed an invention disclosure (patent pending).

Data-Driven AC Microgrid Protection System | *B.Tech Major Project*

Ongoing

PSCAD/EMTDC, MATLAB, Python, Machine Learning

- Developing an ML fault-protection scheme that detects, classifies, and locates AC-microgrid faults from voltage/current waveforms, overcoming fixed-threshold relay limits.
- Built and verified an eight-fault-type simulation model generating labeled waveform data, establishing the pipeline for training adaptive relay decisions.

Intelligent Transformer Health Monitoring using DGA & ML

2026

Python, XGBoost, SHAP, Flask, React, IEC 60599 / IEEE C57.104

- Diagnosed **7 transformer fault types** from DGA with **80% accuracy**, beating the Duval-triangle baseline by **+23 pts** (57%→80%); physics-informed IEC 60599 features, probability calibration (ECE ↓**33%**), and **SHAP** confirming acetylene-arcing signatures.
- Shipped end-to-end: **Flask REST API** with live 7-gas DGA simulator, **React** dashboard with real-time fault visualization, **LLM** diagnostic chatbot with live sensor context, and fleet **risk-ranking** view.

EDUCATION

KIIT University, Bhubaneswar

Jul 2023 – Aug 2027

B.Tech in Electrical Engineering

CGPA: 7.22/10

- Relevant Coursework:** Power System Analysis, Electrical Machines, PLC Programming, Industrial Automation, Power Electronics, Control Systems, Solid State Devices, Signal Processing, Digital Logic.

RESEARCH & ACHIEVEMENTS

- Research Paper (Under Review):** Co-authored “End-to-End TinyML Predictive Maintenance System for Industrial Motors,” submitted to **ICIDeA 2026** (AI/ML & Signal Processing for Power & Energy Applications).
- Research Paper (Under Review):** Co-authored “Digital Twin-Based EV Battery Fault Diagnosis & Protection System with Cloud & CAN Integration,” submitted to **PEDES 2026** (Renewable Energy Systems & Energy Storage).
- Generative AI (Udemy):** Completed two applied GenAI courses — LangChain & Huggingface and Gen AI Apps with Gemini Pro — covering LLMs, RAG pipelines, and end-to-end GenAI application development.
- Smart India Hackathon:** Participated in SIH, contributing to a team solution under a national-level problem statement.